

Articles

Randomised trial of plasma exchange, intravenous immunoglobulin, and combined treatments in Guillain-Barré syndrome

*Plasma Exchange/Sandoglobulin Guillain-Barré Syndrome Trial Group**

Summary

Background The relative efficacy of plasma exchange (PE) and intravenous immunoglobulin (IVIg) for the treatment of Guillain-Barré syndrome has not been established. We compared PE with IVIg, and with a combined regimen of PE followed by IVIg, in an international, multicentre, randomised trial of 383 adult patients with Guillain-Barré syndrome.

Methods The patients were randomly assigned PE (five 50 mL/kg exchanges over 8–13 days), IVIg (Sandoglobulin, 0.4 g/kg daily for 5 days), or the PE course immediately followed by the IVIg course. The inclusion criteria were severe disease (aid needed for walking) and onset of neuropathic symptoms within the previous 14 days. Patients were followed up for 48 weeks.

Findings Four patients were excluded because they did not meet the randomisation criteria. All the remaining 379 patients were assessed for the major outcome criterion—change on a seven-point disability grade scale—by an observer unaware of treatment assignment, 4 weeks after randomisation. At that time, the mean improvement was 0.9 (SD 1.3) in the 121 PE-group patients, 0.8 (1.3) in the 130 IVIg-group patients, and 1.1 (1.4) in the 128 patients who received both treatments (intention-to-treat analysis). None of the differences between the groups for this major outcome criterion was significant. The difference between PE alone and IVIg alone was so small that a 0.5 grade difference was excluded at the 95% level of confidence. There was no significant difference between any of the treatment groups in the secondary outcome measures: time to recovery of unaided walking, time to discontinuation of ventilation, and trend describing the recovery from disability up to 48 weeks. There was a non-significant trend towards a more favourable outcome on some outcome measures with combined treatment.

Interpretation In treatment of severe Guillain-Barré syndrome during the first 2 weeks after onset of neuropathic symptoms, PE and IVIg had equivalent efficacy. The combination of PE with IVIg did not confer a significant advantage.

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Introduction

Plasma exchange (PE) shortened the duration of disability in Guillain-Barré syndrome in comparison with standard supportive treatment in two large, open, controlled trials.^{1,2} In a comparative open trial of intravenous immunoglobulin (IVIg) and PE, the two treatments had similar effects on recovery time.³ IVIg is potentially a safer and more convenient treatment, but it has not been universally accepted as the preferred treatment because of reports of continued disease progression or relapse after IVIg^{4,5} and the fact that the IVIg trial (150 patients)³ was smaller than the PE trials (242¹ and 220²). Although PE (and probably IVIg) shortens the average duration of disease, about 20% of patients are left with substantial disability after either treatment.^{1–3} The mechanisms by which IVIg has a beneficial effect may include anti-idiotypic suppression of autoantibodies.^{6–8} After the removal of immunoglobulin and antibodies by PE, rebound synthesis of antibodies to myelin of peripheral nerves may be stimulated.⁹ The administration of IVIg immediately after PE might prevent this rebound synthesis. We have undertaken a multicentre, randomised controlled trial to find out whether IVIg is equivalent to or superior to PE in the treatment of Guillain-Barré syndrome, and whether PE followed by IVIg is superior to the better single treatment.

Patients and methods

Patients

Patients with Guillain-Barré syndrome were recruited in 38 centres in 11 countries. Inclusion criteria were: diagnosis by a qualified neurologist; satisfaction of accepted clinical and cerebrospinal-fluid diagnostic criteria;¹ severe disease (requiring aid to walk or worse); age over 16 years; and onset of neuropathic symptoms within the previous 14 days. We excluded patients with atypical forms of Guillain-Barré syndrome, serious pre-existing other disease, or contraindications to PE or IVIg.

The protocol was approved by the ethics committees of all centres, and informed consent was obtained from all patients before randomisation. Treatment had to be started within 8 h of randomisation. Each centre was given a set of sealed, sequentially numbered randomisation envelopes, which had been prepared by the trial statistician. At randomisation, the envelope was opened to reveal the assigned treatment. A record was kept of all patients admitted to each centre but not entered into randomisation.

Treatment

PE—patients received five 50 mL/kg plasma exchanges. If an exchange was inadequate, a sixth exchange was given to achieve a total exchange of 250 mL/kg. The exchanges were completed during the 8–13 days after randomisation.

IVIg—patients received human immunoglobulin (Sandoglobulin, Sandoz, Basel, Switzerland) 0.4 g/kg daily for 5

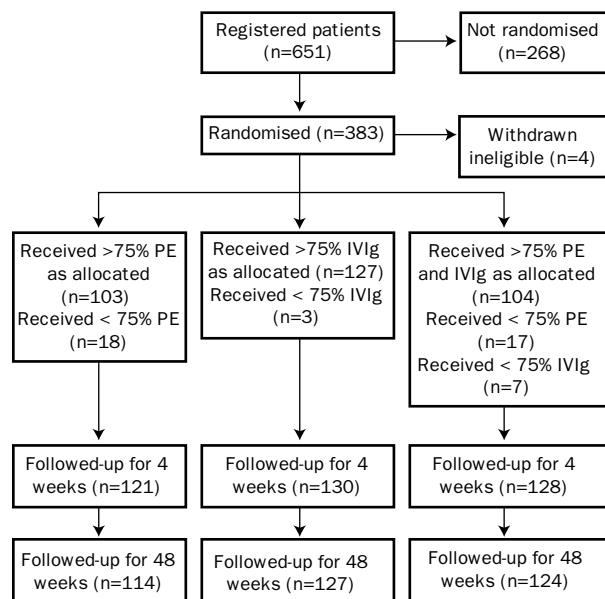


Figure 1: Trial profile

days starting on the day of randomisation (day 1). Sandoglobulin is prepared by fractionation and pH4/pepsin treatment of screened serum in batches pooled from more than 8000 unremunerated donors.

PE followed by IVIg—patients received five 50 mL/kg plasma exchanges starting on the day of randomisation, followed by Sandoglobulin 0.4 g/kg daily for 5 days starting on the day after the last PE.

Other treatment followed the regimen usual in each centre. The use of steroids (except for treatment of intercurrent medical conditions) was discouraged. Retreatment with the original treatment was permitted in the event of a relapse, defined as an increase of 1 or more disability grades for at least 1 week after the disability grade had been stable or had improved by 1 or more grades for at least a week.

Assessments

Disability grade, arm grade, and vital capacity were assessed at randomisation and after 2, 4, 8, 12, 24, and 48 weeks. The presence or absence of sensory deficit during the neurological examination at randomisation was recorded. Neurophysiological assessments were made before or within 5 days of randomisation. The distally evoked compound muscle action potential amplitudes of at least one upper-limb and one lower-limb nerve were expressed as a percentage of the lower limit of normal for the laboratory, and the average of this percentage calculated for each patient.¹⁰ 4 weeks after randomisation, the disability and arm grades and vital capacity were assessed by an examiner who had no knowledge of the treatment.

The disability grades were: 0=healthy, no signs or symptoms of Guillain-Barré syndrome; 1=minor symptoms or signs and able to run; 2=able to walk 5 m across an open space without assistance; 3=able to walk 5 m across an open space with the help of one person and waist-level walking-frame, stick, or sticks; 4=chairbound/bedbound: unable to walk as in 3; 5=requiring assisted ventilation (for at least part of day or night); 6=dead.

The arm grades were: 0=normal; 1=minor symptoms or signs but able to put hand on top of head when sitting with head upright and able to oppose thumb to each fingertip; 2=able to do either of the tasks in 1 but not both; 3=some movements but unable to perform either of the tasks in 2; 4=no movement; 5=dead.

Statistical analysis

The objectives were to establish that IVIg is equivalent to or superior to PE, and to establish that PE followed by IVIg is

	PE (n=121)	IVIg (n=130)	PE+IVIg (n=128)
Male/female	74 (61%)/47 (39%)	79 (61%)/51 (39%)	72 (56%)/56 (44%)
Mean (SD) age in years	51.0 (16.5)	51.5 (17.9)	54.3 (17.2)
Mean (SD) base-line assessment			
Disability grade	3.9 (0.5)	4.0 (0.5)	3.9 (0.6)
Arm grade	2.0 (0.9)	2.0 (0.9)	1.9 (0.9)
Vital capacity	2.3 (1.2)	2.4 (1.4)	2.3 (1.4)
Mean (SD) delay from onset of neuropathy to randomisation in days	6.9 (3.2)	6.4 (3.2)	6.7 (3.2)
Number of patients with characteristic at baseline			
Ventilation required	12 (9.9%)	15 (11.5%)	20 (15.6%)
DCMAP <20% normal*	16 (13.9%)	19 (16.1%)	14 (11.7%)
No sensory symptoms or deficit†	15 (12.6%)	19 (14.8%)	15 (12.0%)
Previous enteritis	29 (24.0%)	24 (18.5%)	25 (19.5%)

DCMAP=distally evoked compound muscle action potentials.

*Information missing for 5 patients in PE group, 12 in IVIg group, and 9 in PE+IVIg group.

†Information missing for 2 patients in PE group, 2 in IVIg group, and 3 in PE+IVIg group.

Table 1: Clinical features of treatment groups at randomisation

superior to the better single treatment. Treatments were taken to be equivalent if the 95% CI of the difference in the mean improvement in disability grade after 4 weeks between the two groups excluded a true difference of more than 0.5 grade. This difference was chosen because we agreed that, although a change of 1.0 grade could be reliably measured and was clinically meaningful, a change of less than 0.5 grade would be clinically insignificant. The SD of the improvement in disability grade at 4 weeks had been 1.2 in the placebo group of a previous trial;¹¹ therefore, three groups of 122 patients were expected to give 90% power to detect a true difference of 0.5 or more in the IVIg versus PE comparison and to exceed 0, given a true difference of 0.5, in the comparison of the combined treatment versus the better of IVIg or PE, on one-sided tests with $p=0.025$. An interim analysis was planned when data on the primary endpoint had been collected for 225 patients. If the mean differences between any of the regimens at that time were greater than 3.5 SDs, a clinical audit committee would decide whether to stop recruitment.

Secondary outcome measures were time from randomisation to unaided walking, time to permanent discontinuation of artificial ventilation, and average rate of recovery derived from the changes in disability grade over the 48-week follow-up period. Other outcome measures of interest were the number of deaths and the proportion of patients who were still unable to walk without aid (grade 3) after 48 weeks.

Results

Randomisation

Between Jan 30, 1993, and April 30, 1995, 383 patients were enrolled and assigned randomised treatment. During the same period, 268 patients were seen at the trial centres but not entered into randomisation (figure 1). The commonest reasons for non-randomisation were disease severity less than grade 3 (70 patients), onset to diagnosis/referral delay more than 14 days (58), other active medical disorders (35), and atypical forms of Guillain-Barré syndrome, such as Miller Fisher syndrome (27). Two patients were excluded from analysis because the diagnosis was later discovered to be incorrect (one

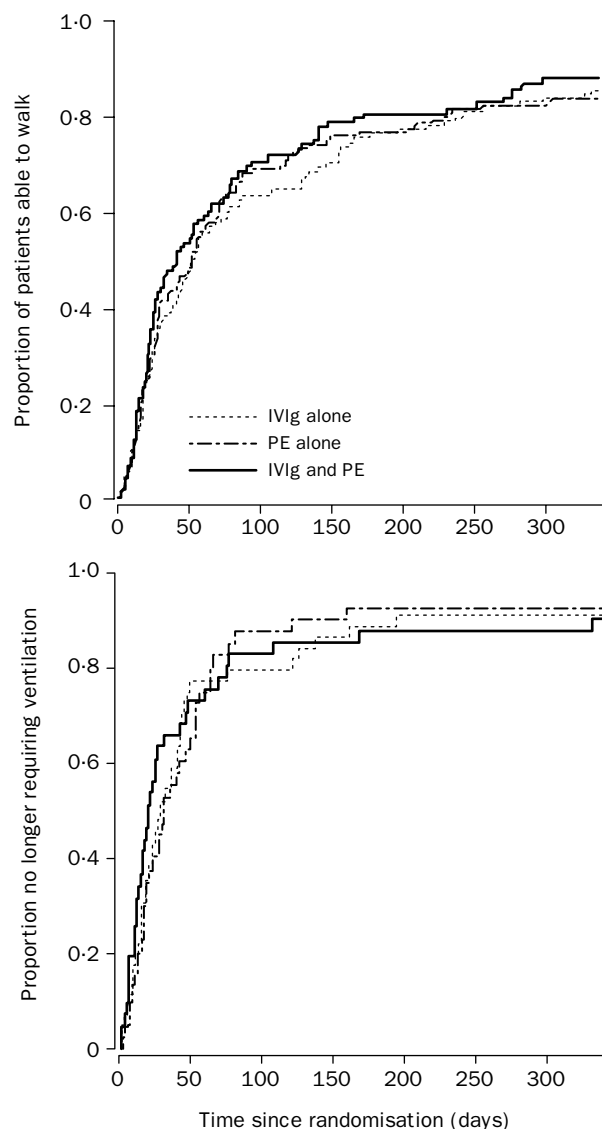


Figure 2: **Survival analysis of time to unaided walking and time to discontinuation of artificial ventilation**

PE, one IVIg group) and two because they had been entered more than 14 days after the onset of neuropathic symptoms (one PE, one PE+IVIg). At randomisation, the three treatment groups were evenly matched for distributions of age, sex, severity of disability, delay from onset, and features known, or suspected, to be related to prognosis (table 1). The majority of patients in all three groups were bedbound and, on average, were entered into randomisation 1 week after the onset of neuropathic symptoms. More than 80% of patients who received PE were treated with continuous-flow rather than intermittent-flow machines. The interim analysis of the disability-grade improvement after 4 weeks for 225 patients showed no differences between the groups sufficient to justify stopping of recruitment.

Major outcome criterion

There were no significant differences between the groups in the major outcome criterion, the mean disability-grade improvement after 4 weeks (table 2). The difference in improvement between the PE group and the IVIg group was only 0.09 grade; the 95% CI of this difference was

−0.23 to 0.42, so the previously defined criterion for equivalence of these two treatments was met. The difference between the PE+IVIg and IVIg alone groups was 0.29 grade (95% CI −0.04 to 0.63), and that between the PE+IVIg and PE alone groups was 0.20 grade (−0.14 to 0.54). Thus, neither was significantly different from zero, and the combined treatment was therefore not superior to either treatment alone. The 95% CI of the differences were slightly too wide to fulfil the criterion of equivalence defined for equivalence of the single treatments.

Secondary outcome criteria

There were no significant differences between the treatment groups in any of the secondary outcome criteria. The time to recover the ability to walk unaided, assessed by survival analysis, did not differ significantly (log-rank test¹⁰) between the three groups (figure 2). The median time to unaided walking was slightly shorter in the PE+IVIg group than in the PE alone or IVIg alone groups, but these differences were not significant (table 2). The proportion of patients still unable to walk after 48 weeks did not differ significantly between the groups (table 2).

The median time from randomisation to discontinuation of ventilation in the patients who needed ventilation at any time was similar in all three groups (figure 2). The mean times to discontinuation of ventilation were also similar among those patients who were required ventilation at the time of randomisation and those who required ventilation subsequently. The proportion of patients who died was higher among those who required ventilation (12% overall; 3/40 in the PE group, 4/44 in the IVIg group, and 8/41 in the PE+IVIg group) than among those who did not require ventilation (4/234 [1.7%]).

The plot of mean grade at each follow-up showed that the pattern of recovery was very similar in all three groups. In an analysis with cubic trend lines (third-degree polynomial cubic curves) fitted to each individual separately, the average trends differed by no more than could be explained by chance (figure 3).

Other outcome measures

During the first 4 weeks, there were two deaths in the PE group (one from hypotension, one from bowel infarction), three in the IVIg group (one each from cardiac arrhythmia, pneumonia, and myocardial infarction), and three in the PE+IVIg group (two from septicaemia, one from pneumonia). The treatment regimen was not thought to have contributed to death in any of these cases. During later follow-up, there were three more deaths in the PE group (one each from intestinal obstruction, pneumonia, and septicaemia), three in the IVIg group (one from asthma, two from pneumonia), and five in the PE+IVIg group (one each from pancreatitis, lung abscess, and pneumonia, and two from carcinoma). The only death judged by the treating neurologist to be attributable to a treatment regimen was in a patient assigned PE alone, who died after 11 weeks from pneumonia that might have been prolonged by the use of PE.

The median times to hospital discharge and to return to work, calculated by survival analysis, did not differ significantly between the treatment groups (table 2).

	PE (n=121)	IVIg (n=130)	PE+IVIg (n=128)
Mean (SD) change in disability grade after 4 weeks	0.9 (1.3)	0.8 (1.3)	1.1 (1.4)
Number ventilated after randomisation	28 (23.1%)	29 (22.3%)	21 (16.4%)
Median (IQR) days to stop artificial ventilation (for all ventilated patients)	29 (14–57)	26 (15–45)	18 (10–56)
Median (IQR) days to unaided walking	49 (19–148)	51 (20–164)	40 (19–137)
Median (IQR) days to hospital discharge	63 (28–124)	53 (21–135)	51 (24–117)
Median (IQR) days to return to work	290 (122–>400)	371 (129–>400)	281 (96–>400)
Number unable to walk unaided after 48 weeks*	19 (16.7%)	21 (16.5%)	17 (13.7%)
Deaths	5 (4.1%)	6 (4.6%)	8 (6.3%)

*Missing for 7 patients in PE group, 1 in IVIg group, and 6 in PE+IVIg group.

Table 2: **Outcome measures**

During the 48 weeks of follow-up, relapses were reported in seven patients in the PE group, four in the IVIg group, and nine in the PE+IVIg group.

Prognostic factors

We did a linear regression analysis of the change in disability grade after 4 weeks, taking into account six factors that may have an effect on outcome—sex, age, time from onset of neuropathic symptoms to randomisation, occupational group, preceding gastrointestinal infection, and arm grade at randomisation, as well as treatment group. In this analysis, sex and occupational group had no significant effect. Treatment group had an effect of borderline significance ($p=0.07$); although the adjusted mean grade change was only 0.1 (95% CI -0.2 to 0.4) more in the PE group than in the IVIg group, it was 0.36 (0.04 to 0.67) more in the PE+IVIg group than in the IVIg alone group. This analysis was repeated for a subset of 344 patients with complete neurophysiological data; the presence of sensory deficit had no significant effect, but the distally evoked compound muscle action potential amplitude did ($p=0.01$). In this analysis, treatment group had a significant effect ($p=0.04$). The difference in mean grade change between the PE and IVIg group was not significant (0.15 [-0.19 to 0.48]), but in the PE+IVIg group, after this adjustment, there was improvement of 0.43 grade (0.09 to 0.77) more than in the IVIg alone group. Cox regression analyses of the time to unaided walking with adjustment for the same prognostic factors gave similar results to those without such adjustment. Recovery was slightly faster in the PE+IVIg group than in either single-treatment group ($p=0.07$) for all the patients and for the subset of 344 patients with complete neurophysiological data when distally evoked compound muscle action potential amplitude was included in the model ($p=0.05$). Analyses that allowed for treatment-centre effects on treatment differences (interactions or effect modification) and rates of recovery showed little or no effect ($p=0.11$ and $p=0.13$, respectively). Furthermore, allowance for treatment centre as a confounder resulted in negligible changes in the treatment-group differences and CIs.

Side-effects

Complications attributed to treatment occurred in eight patients treated with PE alone (hypotension in five,

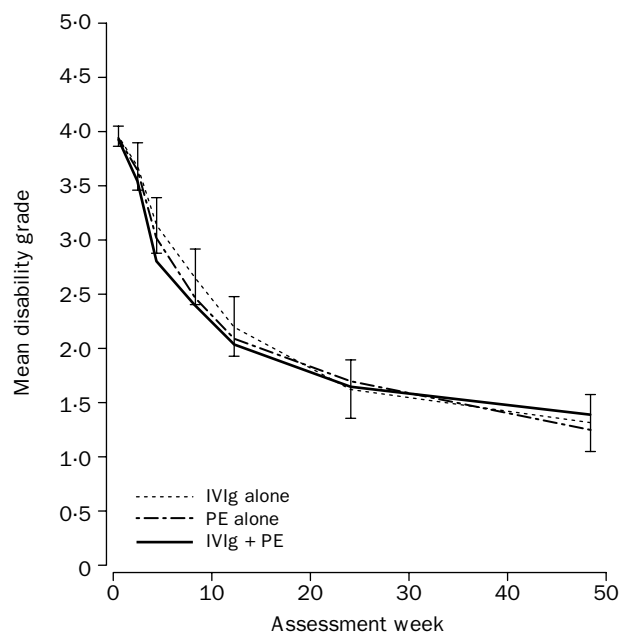


Figure 3: **Trend analysis of recovery of disability grades from randomisation to 48 weeks**

Vertical bars=95% CI of mean for IVIg group.

septicaemia, pneumonia, malaise, abnormal clotting, and hypocalcaemia in one each) and in six patients treated with IVIg alone (nausea or vomiting in two, meningism, exacerbation of chronic renal failure, possible myocardial infarction, and painful erythema at the infusion site in one each). In the PE+IVIg group, six patients had complications attributed to PE (hypotension in three, pulmonary oedema, retroperitoneal haemorrhage, and endocarditis with cerebral embolism in one each) and nine had complications attributed to IVIg (rigor in four, fever in two, flu-like symptoms, malaise with nausea and myalgia, hypotension, chest pain in one each). Some patients had more than one complication. One patient in the PE+IVIg group relapsed and was retreated with combined treatment, which caused an additional complication; a haematoma from the PE catheter required blood transfusion and discontinuation of PE. Treatment had to be curtailed on account of side-effects in three patients treated with PE alone and two patients treated with IVIg alone. In the PE+IVIg group, side-effects led to discontinuation of PE in three patients and IVIg in five patients. The only side-effects with permanent sequelae occurred in the patient who died of pneumonia after having received PE alone and in one patient in the PE+IVIg group, who developed a central-line infection from the PE catheter, and then endocarditis, cerebral embolism, and hemiparesis. Abnormal liver-enzyme activities were reported in 14.9% of the PE group, 8.5% of the IVIg group, and 8.6% of the PE+IVIg group at randomisation and in 16.7%, 16.3%, and 12.5%, respectively, after 2 weeks.

Analysis according to treatment received

Less than 75% of the planned dose was given to 18 patients (13.8%) in the PE group and three (2.3%) in the IVIg group ($p=0.0003$). In the PE+IVIg group less than 75% of the planned amount of PE was given to 17 patients and less than 75% of the planned dose of IVIg was given to seven patients (this number includes two

patients who improved so much after PE that they refused IVIg). Three patients who were assigned to PE alone received IVIg in breach of protocol, in one case because venous access for PE could not be achieved. Two patients who were assigned IVIg received PE in breach of protocol, in both cases at the insistence of the treating physician because of worsening, during IVIg treatment in one and after IVIg treatment in the other. When the analysis of the major outcome criterion was restricted to patients who had received at least 75% of the planned dose, and who had not received non-assigned treatment, the overall conclusion did not change. The mean (SD) grade change after 4 weeks was 0.9 (1.2) in the PE group, 0.8 (1.3) in the IVIg group, and 1.1 (1.2) in the PE+IVIg group.

Discussion

This trial showed that IVIg is equivalent to PE in reducing the amount of disability at 4 weeks after treatment in Guillain-Barré syndrome. This conclusion is strengthened by the absence of significant differences in any of the secondary outcome measures. Moreover, there was no difference in time to hospital discharge or time to return to work. These findings confirm the conclusion of the first comparative trial of PE and IVIg in Guillain-Barré syndrome.³ The results are also consistent with a small pilot trial that compared PE with IVIg.¹² The finding that PE and Sandoglobulin in this trial, and PE and Gammagard (Baxter Healthcare), in Van der Meché and colleagues' trial³ were of equal efficacy does not exclude the possibility that the therapeutic effect is dependent on the method of immunoglobulin purification, since this method was then (but is not now) very similar for these two products.

Our second objective was to show whether a regimen of PE followed by IVIg is superior to the better single treatment. The primary analysis of the reduction in amount of disability after 4 weeks showed a small but not significant advantage for the combined regimen compared with either treatment alone. When this analysis was repeated after significant prognostic factors had been taken into account, the difference between the combined regimen and IVIg alone was just significant, in favour of the combined regimen. The confidence limits of the difference were consistent with either a very small (less than 0.1) or a moderately large (almost 0.8) extra reduction in disability grade. There were non-significant differences in favour of the combined regimen in one of the three secondary outcome measures (median time to unaided walking). There have been no previous studies of combined PE and IVIg in Guillain-Barré syndrome.

Our analysis confirmed reports that older age, small distally evoked compound muscle action potential amplitude, worse arm grade, and previous gastrointestinal infection have an adverse effect on prognosis, irrespective of treatment.^{3,11,13-16} In our study, patients with a longer delay from onset to randomisation had significantly ($p < 0.001$) more improvement after 4 weeks than those with short delays: patients with onset-to-randomisation delays of 12–14 days improved by nearly a whole grade more than patients with delays of less than 2 days. The simple explanation for this finding is that those randomised later were closer to the onset of spontaneous recovery. However, patients with a rapid onset of severe disability have a worse prognosis,⁵⁻⁷ and would have been

more likely to be admitted to hospital and randomised early in the course of their disease.

Distinguishing between side-effects of treatment and complications of the disease can be difficult. There were no significant differences between the groups in the numbers of side-effects attributed by the treating neurologist to the treatments or in the numbers of patients reported at 2 or 4 weeks after randomisation to have infections requiring antibiotic treatment, abnormalities of liver enzymes,¹⁸ or cardiac arrhythmias. PE in seriously ill patients may require the insertion of a central venous catheter and may increase the risk of septicaemia, which caused endocarditis and cerebral embolism in one patient treated with PE alone in our study. PE also removes immunoglobulin and clotting factors, which could in theory increase the risk of infection and haemorrhage, but neither occurred more frequently in the PE than the IVIg group in this trial.

IVIg is a much simpler and more comfortable treatment for the patient than PE. In this trial, less than 75% of the planned IVIg treatment was given in only 2.3% of patients; less than 75% of planned PE was given in 13.8%. In previous trials PE was curtailed in 10%,¹ 16%,³ and 18%² of patients.

On grounds of equal therapeutic benefit, greater convenience, and similar overall cost, IVIg may be preferable to PE for treatment of adult patients with Guillain-Barré syndrome, provided that they are so severely affected that they at least require aid to walk, that the disorder is diagnosed during the first 2 weeks of the illness, and that there are no contraindications to IVIg. This recommendation is offered with two caveats. First, Guillain-Barré syndrome still causes substantial mortality and requires optimum intensive care, preferably in centres with experience of treating acute neuromuscular paralysis. The simplicity of IVIg compared with PE should not tempt centres without appropriate intensive-care facilities to care for these patients. Second, the long-term risks of human blood products, either IVIg or albumin, may not be fully known. The possible slight increase in efficacy with PE before IVIg is not sufficient to justify the greater inconvenience, costs, and risks of the combined regimen. Even with IVIg, 4.6% of patients died and 11.9% were unable to walk without aid after 48 weeks. Thus, the search for superior additional or alternative treatments must continue.

Plasma Exchange/Sandoglobulin Guillain-Barré Syndrome Trial Group

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